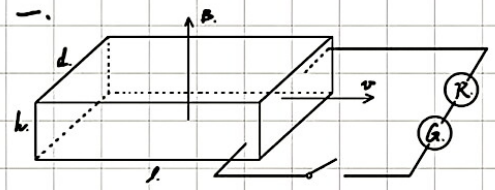
2025年中国科学技术大学科学营一物理试题

一、



解. 不妨假设电解质中电荷密度为n，则电流为

$I = dhvne$ 所受洛伦兹力为 $F_{洛} = IlB = dhvnelB$

$= dhlnevB$

$= qvB$

$\Rightarrow F_{洛} = qvB$

随着长方体容器后部正电荷的积累，则有 $E_q = qvB$

容器类似霍尔元件，正部和后部产生电势差为 $\varepsilon = E\cdot d = vBd$

电解质溶液电阻为 $r = \rho d/hl$

$\Rightarrow I = Bvd \Big/ \left(\frac{\rho d}{hl} + R_g + R\right) \Rightarrow I^{-1} = \frac{1}{Bvd}R + \left(R_g + \frac{\rho d}{hl}\right)\frac{1}{Bvd}$

有 $k = 1/Bvd \Rightarrow B = 1/kvd$

$b = \frac{1}{Bvd}\left(R_g + \frac{\rho d}{hl}\right) \Rightarrow \rho = (Bvdb - R_g)hl/d$

二、解.

由动量守恒和余弦定理：

$(mv)^2 = (mu)^2 + \left(\frac{h}{\lambda}\right)^2 + 2mu\frac{h}{\lambda}\cos\alpha$

由能量守恒有：

$\frac{1}{2}mv^2 + \frac{hc}{\lambda_0} = \frac{1}{2}mu^2 + \frac{hc}{\lambda}$

$\Rightarrow (mu)^2 + \left(\frac{h}{\lambda}\right)^2 + 2mu\frac{h}{\lambda}\cos\alpha = (mu)^2 + 2hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)m$

$\Rightarrow u = \frac{\lambda_0 - \lambda}{\lambda_0\cos\alpha}\cdot c$ （通过量级比较 $\Delta\lambda \gg h/mc$）

$\alpha \in [\frac{\pi}{3}, \frac{\pi}{4}] \Rightarrow u \in [\frac{\lambda_0-\lambda}{\lambda_0}\sqrt{2}\cdot c, \frac{\lambda_0-\lambda}{\lambda_0}2\cdot c]$

三.

1. $\frac{GMm}{r_0^2} = m\frac{v_0^2}{r_0} \Rightarrow M = v_0^2 r_0 / G.$

2. $G\frac{Mm}{r^2} = m\frac{v^2}{r} \Rightarrow v = \sqrt{\frac{r_0}{r}}\, v_0$

3. 由于球壳质量不提供向心力.

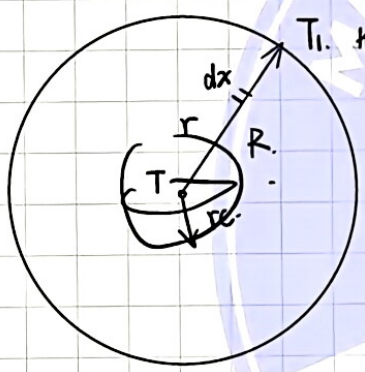
$$\frac{G(M+M')}{r^2} = \frac{v_0^2}{r} \Rightarrow v_0^2 r_0 + GM' = v_0^2 r \Rightarrow G\frac{dM'}{dr} = v_0^2$$

$$M' = \int_{r_0}^{r} 4\pi r^2 \cdot \rho(r)\, dr$$

所以 $\rho(r) = v_0^2 / 4\pi r^2 G.$

$$M' = \int_{r_0}^{11r_0} \frac{v_0^2}{G}\, dr = 10 v_0^2 r_0 / G$$

四.



T_1. 在 (1). $K\Delta T\, 4\pi r^2/dx = \frac{4}{3}\pi r^3 \cdot q.$

$$\frac{W/m^3 \cdot m^2}{W/m\cdot K} = K.$$

$$-3K\, dT\Big|_{T}^{T_1} = q\, r\, dr = \frac{1}{2} q\, dr^2 \Big|_0^R$$

$$\left|3K(T-T_1)\right| = \frac{1}{2} q R^2.$$

$$\frac{4}{3}\pi r^3 q = k\, 4\pi r^2 \cdot dT$$

$$T = \frac{qR^2}{6K} + T_1 \quad \#.$$

(2) $r < R_c$ $\quad -K\, dT\, 4\pi r^2/dr = \frac{4}{3}\pi r^3 \cdot q$

$\Rightarrow 3K(T - T_c) = \frac{1}{2} q R_c^2 \qquad T = T_c + \frac{qR_c^2}{6K}.$

$r > R_c$ $\quad \frac{4}{3}\pi R_c^3 \cdot q = -K\, dT\, 4\pi r^2/dr$

$$-\frac{qR_c^3}{3Kr^2}\, dr = -dT$$

$$-\frac{qR_c^3}{3K} r^{-1}\Big|_{R_c}^{R} = T_c - T_1$$

$$\frac{qR_c^3}{3K}\left(\frac{1}{R_c} - \frac{1}{R}\right) + T_1 = T_c. = T_1 + \frac{qR_c^{\cancel{3}\,2}}{3K}\left(\frac{R-R_c}{R\cancel{R_c}}\right). = T_1 + \frac{qR_c^2}{3K}\frac{R-R_c}{R}.$$

$$T = T_1 + \frac{qR_c^2}{3K}\left(\frac{R}{2R} + \frac{2R-2R_c}{2R}\right)$$

$$= T_1 + \frac{qR_c^2}{3K}\left(\frac{3R-2R_c}{2R}\right) \quad \#$$

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